Eddy Solutions of the Navier-Stokes Equations

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We have recently noticed a class of explicit solutions to the Navier-Stokes equations, periodic in both space variables. The solutions are exponentially decaying and self-similar in the sense that the stream lines are constant in time. Figure 1 represents the stream lines in one cell of one particularly striking flow. The class of solutions is a result of an observation, Lemma 1, concerning eigenfunctions of the Stokes operator with corresponding zero pressure. Through a survey paper [1] of R. Berker, we were led to a 1923 paper [2] of G.I. Taylor in which he remarks on this class. (We would like to thank R. Finn for informing us of [1]). Taylor considered and sketched a simple example which, for historical interest, we have included as figure 2. The complex flows we noticed are a result of special eigenfunctions which are described below.

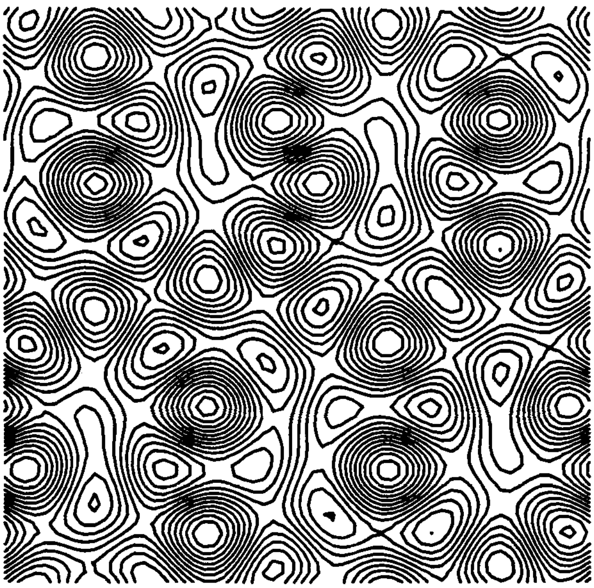

Figure 1: $\psi = (1/4)\cos(3x)\sin(4y) - (1/5)\cos(5y) - (1/5)\sin(5x)$

Consider the Navier-Stokes equations in a simply connected domain Ω. That is, for $\mathbf{x} \in \Omega$ and $t > 0$,

$$\mathbf{u}_t + \mathbf{u} \cdot \nabla \mathbf{u} = \nu \triangle \mathbf{u} - \nabla p \ , \quad \nabla \cdot \mathbf{u} = 0, \tag{1}$$

with $\mathbf{u} = \mathbf{a}$ initially.

Lemma 1 *Suppose* $\mathbf{a}$ *satisfies*

$$\triangle \mathbf{a} = \lambda \mathbf{a} \ , \ \nabla \cdot \mathbf{a} = 0 \text{ in } \Omega. \tag{2}$$

Then $\mathbf{u} = e^{\nu\lambda t}\mathbf{a}$ *satisfies (1) with pressure* p *such that* $\nabla p = -\mathbf{u} \cdot \nabla \mathbf{u}$.

Proof: $\mathbf{u}$ is divergence free since $\mathbf{a}$ is. Also, $\mathbf{u}_t = \nu\lambda\mathbf{u} = \nu \triangle \mathbf{u}$ and hence we need only show that the nonlinear term is a gradient. This is equivalent to showing

$$\frac{\partial}{\partial y}\left(u_1\frac{\partial u_1}{\partial x} + u_2\frac{\partial u_1}{\partial y}\right) = \frac{\partial}{\partial x}\left(u_1\frac{\partial u_2}{\partial x} + u_2\frac{\partial u_2}{\partial y}\right),$$

which follows straightforwardly from (2).

We remark that $\mathbf{u}$ also satisfies the (time dependent) Stokes equations, the linearization of (1), with $p \equiv 0$.

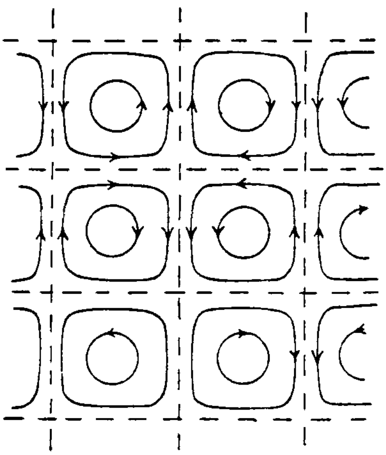

Figure 2: $\psi = \cos(x)\cos(y)$, (taken from [2], pg. 674)

The periodic flows are best described through their stream function. If ψ is an eigenfunction of the Laplacian, eigenvalue λ, then $\mathbf{a} = (\psi_y, -\psi_x)$ satisfies (2), same λ, and $e^{\nu\lambda t}\psi$ is the stream function of the associated Navier-Stokes (or Stokes) flow. Consider eigenfunctions in the case of the 2π-periodic box. All eigenvalues λ are of the form $\lambda = -(n^2 + m^2)$ where n and m are non-negative integers. Given n and m,

$$\cos(nx)\cos(my), \ \sin(nx)\sin(my), \ \cos(nx)\sin(my), \ \sin(nx)\cos(my)$$

are linearly independent eigenfunctions. Define linear combinations of these as (n,m)-eigenfunctions. Simple flows, such as Taylor's, consisting of eddies arranged in regular